CHAPTER 12

# Perfect Competition

ECON101 - Prince Sultan University

OVERVIEW

# Learning objectives

In some markets, so many firms sell the same product that none of them can affect its price. This chapter looks at how a firm in such a market decides how much to produce, and at the profit or loss it ends up with in the short run.

1. Define perfect competition and explain why a firm in a perfectly competitive market is a price taker. NOT price makers
2. Explain how a perfectly competitive firm chooses the output that maximizes its profit.
3. Explain when the firm shuts down temporarily, and describe the three possible short-run outcomes for its profit.

# Perfect competition

**Perfect competition** is a market in which:

- many firms sell identical products to many buyers;
- nothing restricts entry into the market; → free entry, free exit
- firms already in the market have no advantage over new ones;
- sellers and buyers are well informed about prices.

No real market meets all four conditions exactly, but many come close. In Saudi Arabia, examples include:

- date farms in Al-Qassim and Al-Ahsa, and vegetable farms around Riyadh;
- fishing boats selling their catch at the fish markets of Jeddah and Dammam;
- neighborhood laundries, house painters and car-wash services.

# How perfect competition arises

Perfect competition is likely when a market has two features:

### Room for many firms

A single firm reaches its lowest possible cost per unit at an output that is tiny compared with the quantity the whole market buys. The market can therefore hold a large number of firms, each with a small share.

One date farm's harvest is a very small part of all the dates traded in Al-Qassim.

### Nothing to tell the products apart

No firm's product has unique features, so buyers do not care which firm they buy from.

To a buyer, a box of Sukkari dates from one farm is as good as a box of the same variety from the next farm.

# Firms are price takers

A **price taker** is a firm that cannot influence the market price, because its output is a tiny part of the total quantity sold.

Al-Bustan Farm sells boxes of dates at the Buraydah date market. Hundreds of farms offer the same variety there, and every buyer knows the going price: **60 SAR a box**.

### Asks 62 SAR

Buyers simply walk to another stall and pay 60 SAR. The farm sells nothing.

### Asks 58 SAR

The farm sells out at once, but it gives away 2 SAR on every box.

### Charges 60 SAR

The farm sells as many boxes as it wants. Its best choice is to **take the market price**.

# Economic profit and revenue

Each firm aims to make its economic profit as large as possible:

**Economic profit = total revenue – total cost**

- **Total cost** (*TC*) is the opportunity cost of production. It includes **normal profit**, the return an entrepreneur can expect to earn on average. A firm with zero economic profit still earns normal profit.
- **Total revenue** (*TR*) is the price of the output times the quantity sold: $TR = P \times Q$.
- **Marginal revenue** (*MR*) is the change in total revenue when the quantity sold rises by one unit: $MR = \Delta TR / \Delta Q$.

# Running example: Al-Bustan Farm

Al-Bustan Farm grows dates in Al-Qassim. In the short run its palms and land are fixed and cost **40 SAR a day** (total fixed cost). Each day it chooses how many boxes to sell, from 0 to 11, at the market price of **60 SAR a box**.

| Quantity (*Q*, boxes) | Price (*P*, SAR per box) | Total revenue (*TR*, SAR) | Marginal revenue (*MR*, SAR per extra box) |
|---|---|---|---|
| 6 | 60 | 360 | |
| | | | 60 |
| 7 | 60 | 420 | |
| | | | 60 |
| 8 | 60 | 480 | |

- Each extra box adds 60 SAR to total revenue, so ***MR = P***.
- The farm can sell any quantity at 60 SAR, so the demand for its dates is a **horizontal line at the market price**. That line is also its *MR* curve.
- This demand is **perfectly elastic**, because one farm's box is a perfect substitute for another farm's box. The market demand for dates is not perfectly elastic, because other foods are only imperfect substitutes for dates.

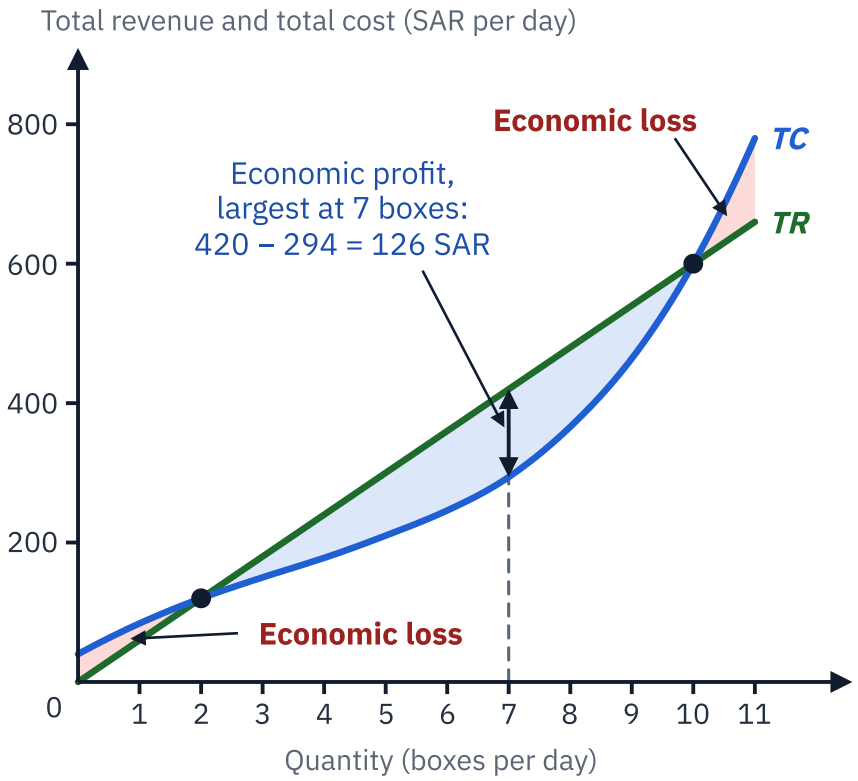
# Choosing output: total revenue and total cost

SAR per day, at a price of 60 SAR a box; economic profit = *TR* – *TC*

| Quantity (boxes) | Total revenue (*TR*) | Total cost (*TC*) | Economic profit |
|---|---|---|---|
| 0 | 0 | 40 | –40 |
| 1 | 60 | 84 | –24 |
| 2 | 120 | 120 | 0 |
| 3 | 180 | 150 | 30 |
| 4 | 240 | 178 | 62 |
| 5 | 300 | 210 | 90 |
| 6 | 360 | 246 | 114 |
| **7** | **420** | **294** | **126** |
| 8 | 480 | 366 | 114 |
| 9 | 540 | 464 | 76 |
| 10 | 600 | 600 | 0 |
| 11 | 660 | 780 | –120 |

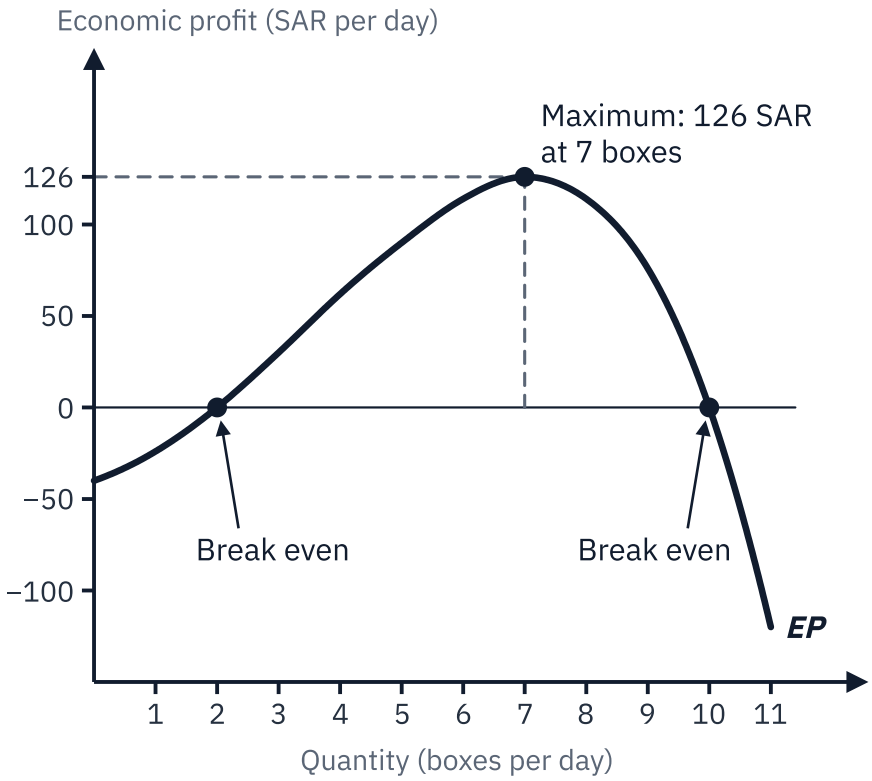
- A price taker chooses only its output. Al-Bustan Farm looks for the quantity that makes *TR* – *TC* as large as possible.
- At 0 and 1 box, revenue is too small to cover total cost, so the farm makes an **economic loss**.
- At 2 and at 10 boxes, *TR* = *TC*: the farm **breaks even**. In between, it makes an **economic profit**.
- Profit is largest at **7 boxes**: 420 – 294 = **126 SAR** a day.
- At 11 boxes, cost has overtaken revenue again.

# Total revenue and total cost curves




- The *TR* curve is a **straight line** from the origin with a slope of 60 SAR: each box adds the market price to revenue.
- The *TC* curve starts at 40 SAR, the fixed cost. It flattens a little over the first few boxes, then rises more and more steeply as diminishing returns set in.
- Where *TR* lies **above** *TC*, between 2 and 10 boxes, the vertical gap is an economic profit. Where it lies **below**, the gap is an economic loss.
- The profit-maximizing output is where the gap is widest: **7 boxes**.

# The economic profit curve




The *EP* curve plots *TR* – *TC* at each output.

- **Low output:** losses. Revenue does not cover total cost; with no boxes at all, the loss is the whole 40 SAR fixed cost.
- **Middle output:** profits, peaking at 126 SAR with 7 boxes.
- **High output:** losses again. Diminishing returns make each extra box more costly, so cost rises faster than revenue.
- The curve crosses zero at 2 and 10 boxes, the **break-even points** at this price.

THE FIRM'S OUTPUT DECISION

# Marginal analysis

A second way to find the best output compares what one more box adds to revenue with what it adds to cost. Marginal revenue stays at the price, while marginal cost eventually rises, so profit is largest where ***MR = MC*** and *MC* is rising.

### If *MR* > *MC*

**Produce more.** The extra box adds more to revenue than to cost, so **profit rises** with output. From 6 to 7 boxes, *MR* = 60 > *MC* = 48, and profit grows by 12 SAR, from 114 to 126.

### If *MR* < *MC*

**Produce less.** The extra box adds more to cost than to revenue, so **profit falls** if output rises. From 7 to 8 boxes, *MR* = 60 < *MC* = 72, and profit drops by 12 SAR, from 126 to 114.

### If *MR* = *MC*

**Best output.** Moving output in **either direction** lowers profit, so profit is at its **maximum**. On the farm's smooth *MC* curve, *MC* = 60 SAR at 7 boxes.

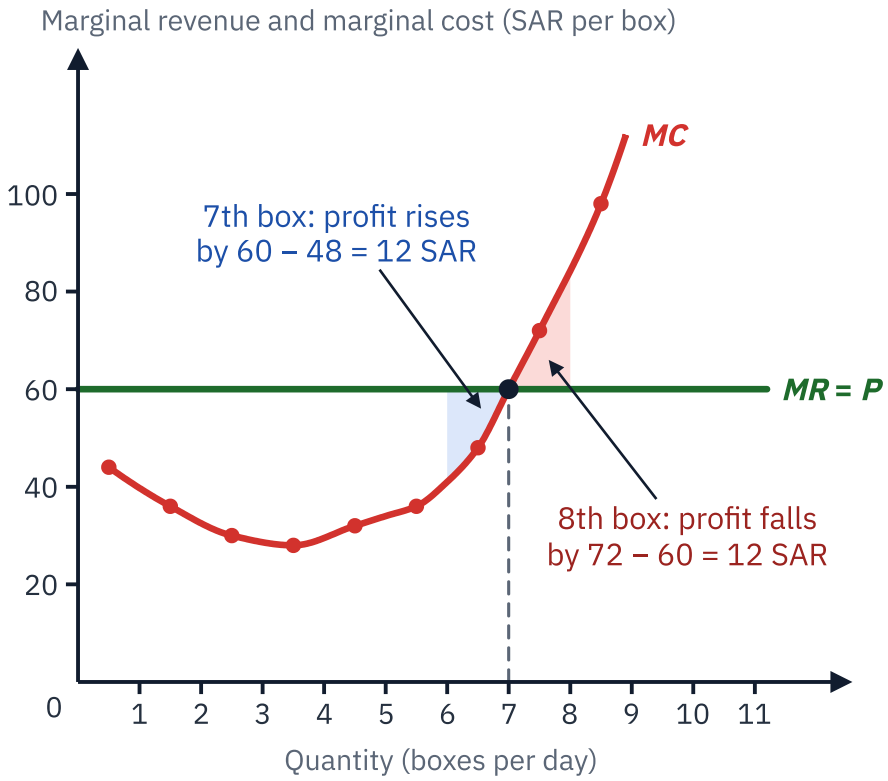
# Al-Bustan Farm: marginal revenue and marginal cost

SAR, at a price of 60 SAR a box; *MR* and *MC* per extra box

| Quantity (boxes) | *TR* | *MR* | *TC* | *MC* | Economic profit |
|---|---|---|---|---|---|
| 5 | 300 | | 210 | | 90 |
| | | 60 | | 36 | |
| 6 | 360 | | 246 | | 114 |
| | | 60 | | 48 | |
| **7** | **420** | | **294** | | **126** |
| | | 60 | | 72 | |
| 8 | 480 | | 366 | | 114 |
| | | 60 | | 98 | |
| 9 | 540 | | 464 | | 76 |

- *MR* and *MC* sit between the two rows they link. From 6 to 7 boxes, revenue rises by 60 SAR and cost by 294 – 246 = 48 SAR.
- While *MR* exceeds *MC*, one more box raises profit: the 6th box adds 24 SAR and the 7th adds 12 SAR.
- The 8th box would cost 72 SAR and bring in only 60, so it would cut profit by 12 SAR.
- So the farm stops at **7 boxes**, with a profit of 126 SAR.

# Profit-maximizing output: *MR* = *MC*




- The *MR* line is horizontal at the market price, 60 SAR.
- Each *MC* value is plotted halfway between the two outputs it links, and the *MC* curve passes through these points.
- **Blue area:** the 7th box brings in 60 SAR and costs 48 SAR, so it raises profit.
- **Red area:** an 8th box would cost 72 SAR and bring in 60 SAR, so it would lower profit.
- Profit is largest where the rising *MC* cuts the *MR* line: **7 boxes**.

THE FIRM'S OUTPUT DECISION

# Output and the law of supply

The profit-maximizing output is the **quantity the firm supplies** at that price. Where marginal cost is rising, the farm keeps adding boxes while the next box costs less than the price, and stops before the first box that would cost more than it brings in.

### 45 SAR → 6 boxes

The 6th box costs 36 SAR, less than the price. The 7th would cost 48 SAR.

### 60 SAR → 7 boxes

The 7th box costs 48 SAR, less than the price. The 8th would cost 72 SAR.

### 85 SAR → 8 boxes

The 8th box costs 72 SAR, less than the price. The 9th would cost 98 SAR.

**A higher price makes more boxes worth producing, so the quantity supplied increases: the firm's behavior follows the law of supply.**

TEMPORARY SHUTDOWN

# When even the best output makes a loss

At a low price, the most profitable output may still bring an economic loss. The firm's response depends on how long it expects the loss to last:

### A lasting loss

The firm leaves the market.

### A temporary loss

The firm stays in the market and either produces or shuts down for a while, **whichever loses less**.

**Economic loss = *TFC* + *TVC* − *TR* = *TFC* + (*AVC* − *P*) × *Q***

- A firm that shuts down produces $Q = 0$. It earns no revenue but must still pay its fixed cost, so its loss equals *TFC*: **40 SAR a day** for Al-Bustan Farm.
- So in the short run, a firm never has to lose more than its total fixed cost.

# Al-Bustan Farm's cost schedule (SAR)

| Quantity (boxes) | *TVC* | *TC* | *AVC* | *ATC* | *MC* (per extra box) |
|---|---|---|---|---|---|
| 0 | 0 | 40 | | | |
| | | | | | 44 |
| 1 | 44 | 84 | 44 | 84 | |
| | | | | | 36 |
| 2 | 80 | 120 | 40 | 60 | |
| | | | | | 30 |
| 3 | 110 | 150 | 36.7 | 50 | |
| | | | | | 28 |
| 4 | 138 | 178 | 34.5 | 44.5 | |
| | | | | | 32 |
| 5 | 170 | 210 | **34** | 42 | |
| | | | | | 36 |
| 6 | 206 | 246 | 34.3 | **41** | |
| | | | | | 48 |
| 7 | 254 | 294 | 36.3 | 42 | |
| | | | | | 72 |
| 8 | 326 | 366 | 40.8 | 45.8 | |
| | | | | | 98 |
| 9 | 424 | 464 | 47.1 | 51.6 | |
| | | | | | 136 |
| 10 | 560 | 600 | 56 | 60 | |
| | | | | | 180 |
| 11 | 740 | 780 | 67.3 | 70.9 | |

- *AVC* = *TVC* ÷ *Q* and *ATC* = *TC* ÷ *Q*, rounded to one decimal place.
- *AVC* is lowest at **5 boxes (34 SAR)** and *ATC* at **6 boxes (41 SAR)**.
- Each *MC* value is the cost of one extra box, so it sits between two rows: the 5th box costs 170 – 138 = 32 SAR.
- The *MC* curve on the graphs is a smooth illustration through these values. It crosses *AVC* and *ATC* at their lowest points.

TEMPORARY SHUTDOWN

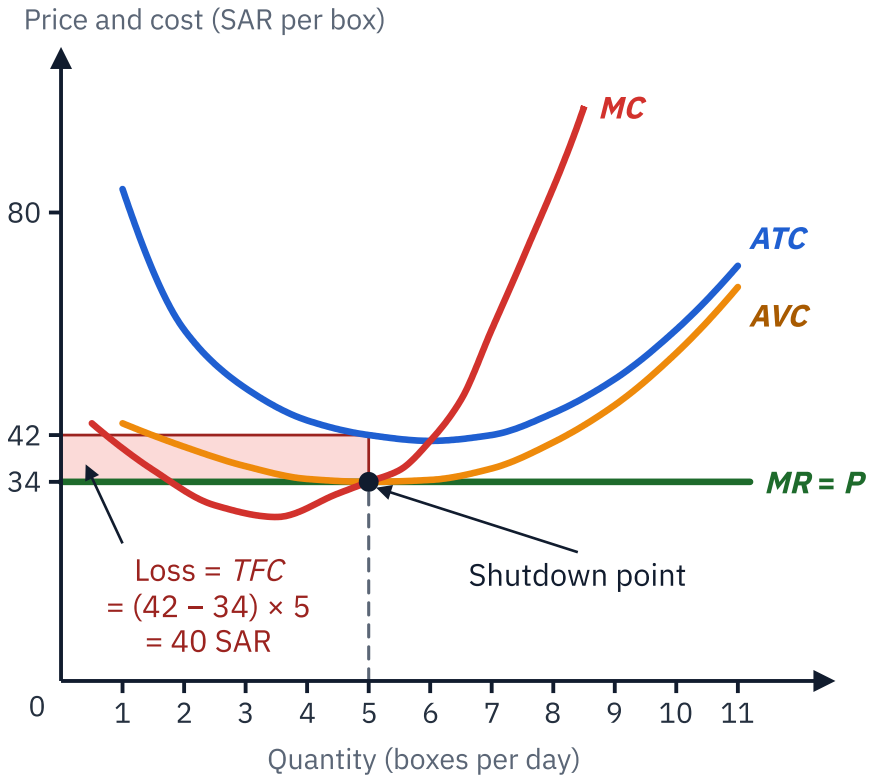
# The shutdown point

The **shutdown point** is the price and quantity at which the firm is indifferent between producing its profit-maximizing output and shutting down. It is at **minimum *AVC***, where the *MC* curve crosses the *AVC* curve, and there the firm's loss equals *TFC*.

| Price (SAR per box) | Best output if it produces | *TR* (SAR) | *TC* (SAR) | Loss if producing (SAR) | Decision |
|---|---|---|---|---|---|
| 38: above minimum *AVC* | 6 boxes | 228 | 246 | 18 | Produce: 18 SAR < 40 SAR |
| 34: equal to minimum *AVC* | 5 boxes | 170 | 210 | 40 | Indifferent: 40 SAR = 40 SAR |
| 33: below minimum *AVC* | 5 boxes | 165 | 210 | 45 | Shut down: lose only 40 SAR |

- At 38 SAR, revenue of 228 SAR covers the whole variable cost of 206 SAR and part of the fixed cost, so producing beats shutting down.
- At 33 SAR, the price is below *AVC* at every output, so revenue never covers even the variable cost: every output loses more than 40 SAR.

# The shutdown point on a graph




- At a price of 34 SAR, *MR* = *MC* on the rising part of *MC* at **5 boxes**, the lowest point of the *AVC* curve.
- Revenue, 34 × 5 = 170 SAR, just pays the variable cost, so nothing is left for the fixed cost.
- **Red rectangle:** the loss, (*ATC* – *P*) × *Q* = (42 – 34) × 5 = 40 SAR, equal to *TFC*.
- At any lower price, producing would lose more than 40 SAR, so the farm shuts down.

# Profit or loss in the short run

The market price comes from market demand and market supply, so it changes when market conditions change. The highest profit the firm can reach is **not always positive**.

To find out which case applies, compare the price with *ATC* at the profit-maximizing output:

## 41 SAR

6 boxes, *ATC* = 41 SAR.
***P* = *ATC*: break even** (normal profit).
*TR* = *TC* = 41 × 6 = 246 SAR: zero economic profit.

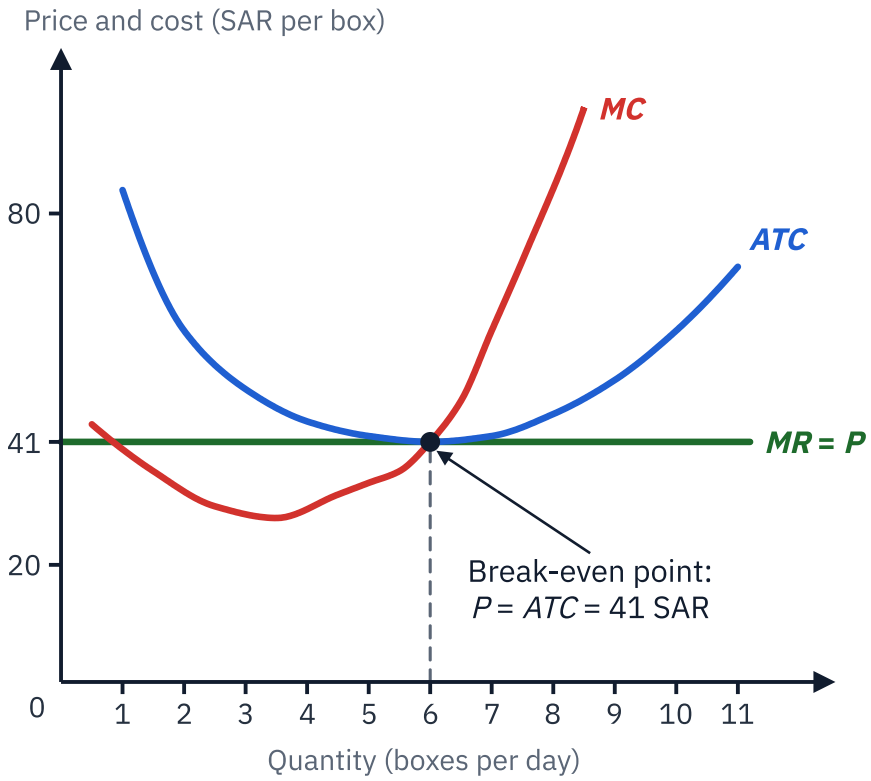
## 60 SAR

7 boxes, *ATC* = 42 SAR.
***P* > *ATC*: economic profit**.
Profit = (60 – 42) × 7 = 126 SAR.

## 34 SAR

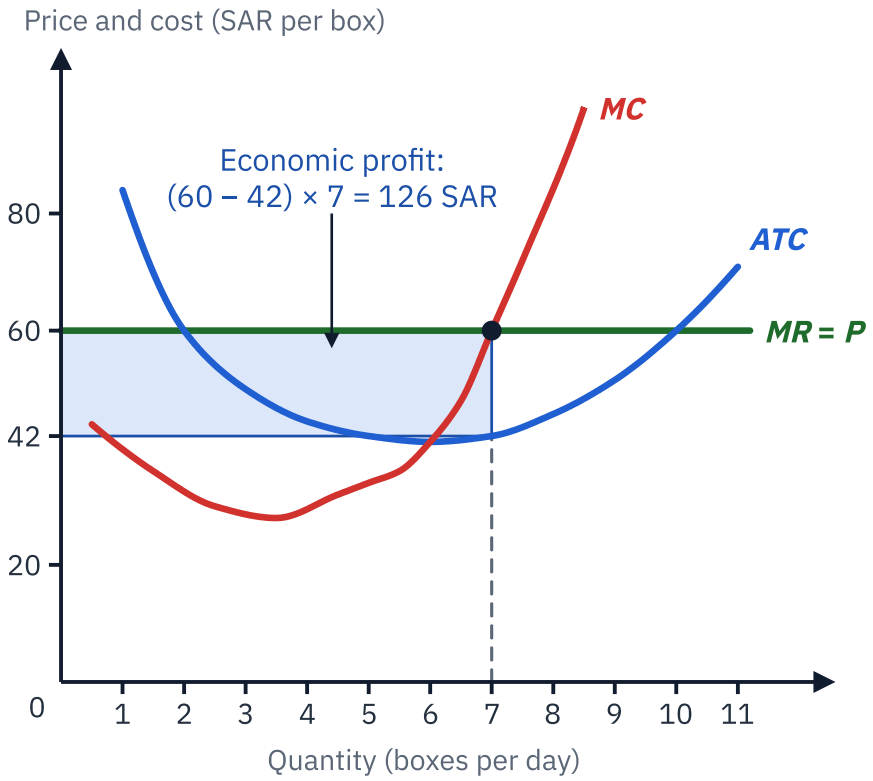
5 boxes, *ATC* = 42 SAR.
***P* < *ATC*: economic loss**.
Loss = (42 – 34) × 5 = 40 SAR.

# Break even: price 41 SAR



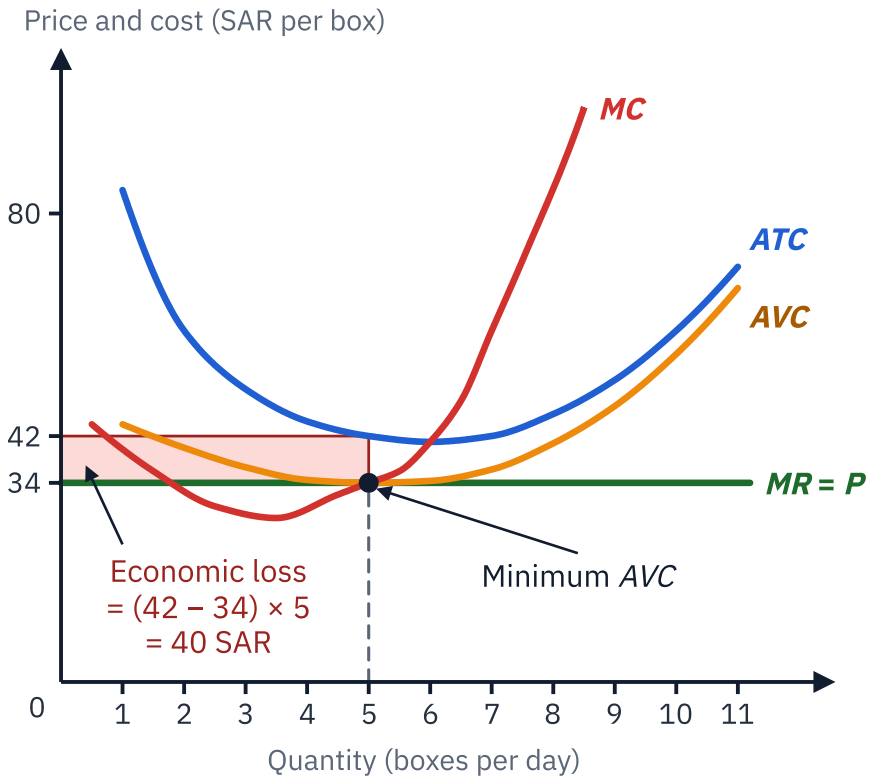

- At 41 SAR, *MR* = *MC* on the rising part of *MC* at **6 boxes**, where *ATC* is at its minimum of 41 SAR.
- Price equals average total cost, so *TR* = *TC* = 41 × 6 = 246 SAR.
- Economic profit is **zero**: the farm **breaks even**.
- Because total cost includes normal profit, the owner still earns **normal profit**.

# Economic profit: price 60 SAR




- At 60 SAR, *MR* = *MC* at **7 boxes**, where *ATC* is 42 SAR.
- Price exceeds average total cost, so each box earns 60 – 42 = 18 SAR more than it costs on average.
- **Blue rectangle:** economic profit = (*P* – *ATC*) × *Q* = 18 × 7 = **126 SAR** a day, the same maximum found from the total curves.

# Economic loss: price 34 SAR




- At 34 SAR, *MR* = *MC* on the rising part of *MC* at **5 boxes**, where *ATC* is 42 SAR.
- Price is below average total cost, so each box costs 8 SAR more than it brings in, on average.
- **Red rectangle:** economic loss = (*ATC* – *P*) × *Q* = 8 × 5 = **40 SAR** a day.
- This price is also the minimum of *AVC*, so the farm is at its shutdown point: the loss equals *TFC*.

SUMMARY

# Key takeaways

- In perfect competition, many firms sell identical products, so each firm is a **price taker**. The demand for its output is horizontal at the market price, and $MR = P$.
- The firm maximizes economic profit, $TR - TC$, by producing the output at which ***MR* = *MC***: 7 boxes for Al-Bustan Farm at 60 SAR.
- At a higher price, more units are worth producing, so the quantity supplied rises: the **law of supply**.
- Facing a temporary loss, the firm produces if the price is above minimum *AVC* and shuts down if it is below. Its short-run loss never has to exceed *TFC*.
- In the short run, the firm **breaks even** if $P = ATC$, makes an **economic profit** if $P > ATC$, and makes an **economic loss** if $P < ATC$.

SOURCE

# Source

Michael Parkin (2022) *Microeconomics, Global Edition*, 14th Edition, published by Pearson Education.